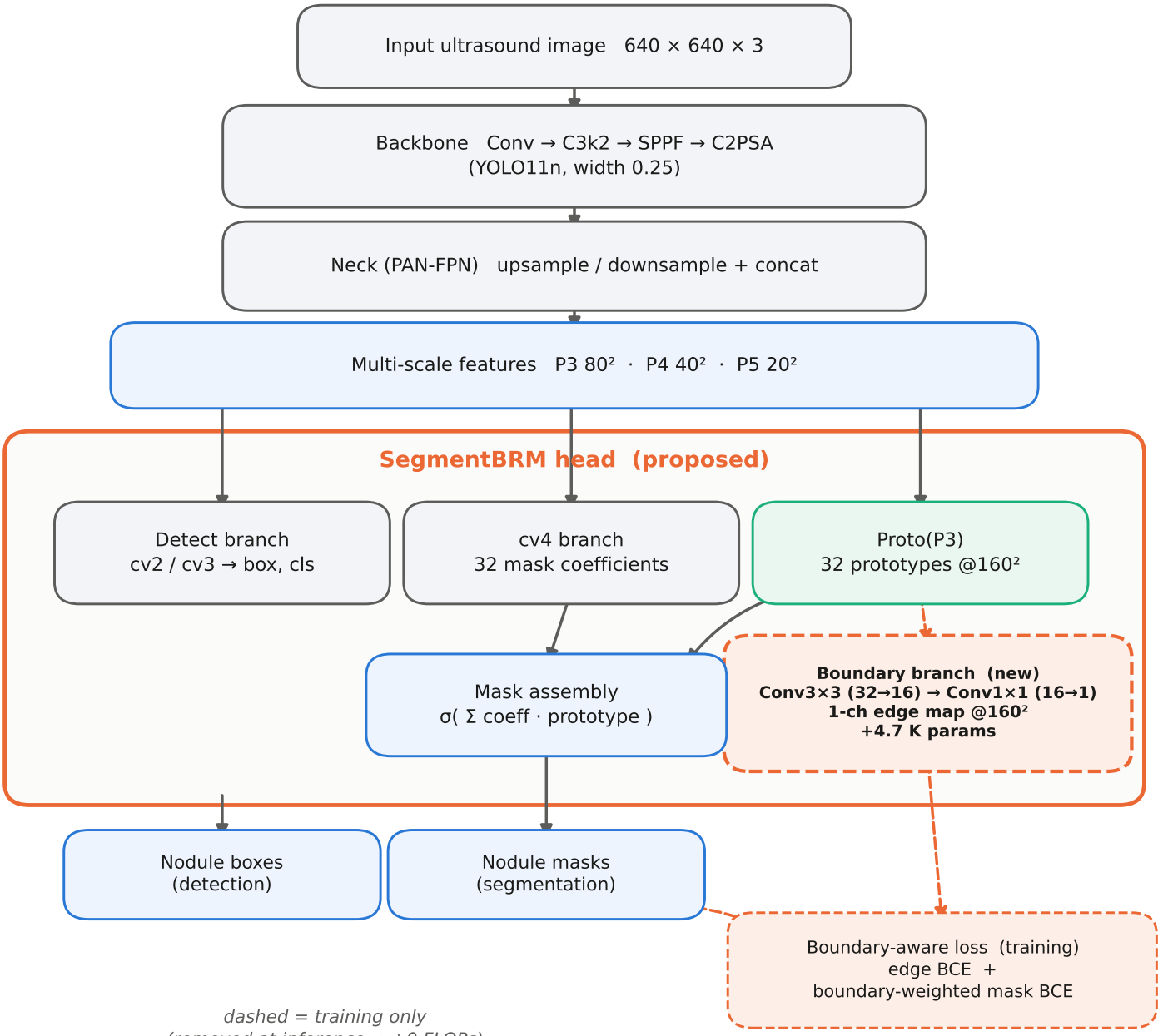
Proposed architecture: YOLO11n-seg with Boundary Refinement Module
Input ultrasound image 640 × 640 × 3
Backbone Conv → C3k2 → SPPF → C2PSA
(YOLO11n, width 0.25)
Neck (PAN-FPN) upsample / downsample + concat
Multi-scale features P3 80² · P4 40² · P5 20²
SegmentBRM head (proposed)
Detect branch
cv2 / cv3 → box, cls
cv4 branch
32 mask coefficients
Proto(P3)
32 prototypes @160²
Mask assembly
σ(Σ coeff · prototype)
Boundary branch (new)
Conv3×3 (32→16) → Conv1×1 (16→1)
1-ch edge map @160²
+4.7 K params
Nodule boxes
(detection)
Nodule masks
(segmentation)
Boundary-aware loss (training)
edge BCE +
boundary-weighted mask BCE
dashed = training only
(removed at inference → +0 FLOPs)